Sample Combining Validation: Truth p_T Spectra (signal + background)

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Abstract

We validate that the 8 SI/DI MC sample pairs (photon5/10/20 signal, jet8/12/20/30/40 background; each with `_nom` + `_double` = 16 per-period files) combine correctly into the signal and jet-inclusive outputs used by the PPG12 efficiency pipeline. Two diagnostics are performed on both crossing-angle periods (0 mrad and 1.5 mrad) and, where applicable, the all-range cross-period merge: (A) per-sample truth photon p_T ratios to the combined signal; (B) per-sample truth leading-jet p_T ratios to the jet-inclusive combined, with explicit zoom on each sample boundary. The combining arithmetic (`hadd` of pre-weighted per-sample files) is exact at floating-point precision for both signal and background in both periods, and the all-range cross-period merge is bit-exact. **However, the combined 1.5 mrad jet spectrum shows a visible step at the 14 GeV sample boundary.** Root cause: the jet8 SI MC was generated with an anomalously narrow truth-vertex distribution (σ_z = 16 cm vs 65 cm for every other sample), and the 1.5 mrad truth-vertex reweight has a sharp peak at $z = 0$ (max weight 4.4×); the two together inflate jet8 SI at 1.5 mrad by a factor of ~ 2.5 relative to every other sample. The prediction $0.591 \times (2.948/1.192) = 1.462$ matches the measured period ratio 1.466 to 0.3%. The fix is upstream — regenerate jet8 SI with the standard wide vertex distribution — not in the combining step itself.

1 Summary

Verdict: Combining arithmetic PASS; 1 warning on per-sample normalization at 14 GeV in 1p5mrad.

- **Signal:** $\sum_{i=1}^{6}$ photon$X_{\text{nom,double}}$ / combined = 1.000000 (max per-bin deviation 2.2×10^{-16}, floating-point). **PASS**.
- **Background:** $\sum_{i=1}^{10}$ jet$X_{\text{nom,double}}$ / combined = 1.00000000 (max per-bin deviation $< 3 \times 10^{-8}$). Combining arithmetic **PASS**.
- **Cross-period additivity:** (0 mrad combined) + (1.5 mrad combined) = all-range combined, bin-by-bin to machine precision. **PASS**.
- **Background jet spectrum shape at 14 GeV (1.5 mrad):** a factor-3.3 step down between $[13.5, 14.0]$ GeV (1.68×10^{11}) and $[14.0, 14.5]$ GeV (5.09×10^{10}) — much larger than the $\sim 0.8\times$-per-0.5-GeV smooth-spectrum expectation. Boundaries at 21, 32, 42 GeV (both periods) and 14 GeV (0 mrad) are within the expected smooth falloff.

Root cause of the 14 GeV 1p5mrad step: the jet8 SI MC has an anomalously narrow truth-vertex distribution (σ_z = 16 cm vs 65 cm for every other sample). Combined with the 1.5 mrad truth-vertex reweight peak (4.4× at $z = 0$), this gives jet8 SI a mean reweight of 2.95 at 1.5 mrad (vs 1.00 for all other samples). The predicted period-ratio for jet8_nom becomes $0.591 \times 2.47 = 1.462$, matching the measured 1.466 to 0.3%. Details and mitigation in Section 5.
Documentation note: the bare (no-period-suffix) per-sample files `MC_efficiency_{photon,jet}X_{nom,double}_`
on disk are stale copies of the 1.5 mrad content (Apr 20 vs the Apr 21 re-run). They are not used

by `merge_periods.sh`, which combines at the per-period *combined* level; this validation reflects that fact.

2 Method

2.1 Inputs

All inputs are per-period MC outputs from `RecoEffCalculator_TTreeReader.C`, with the per-event weight $w = \sigma \times f_{\text{mix}} \times (L/L_{\text{target}}) \times w_{\text{truth vtx}}$ *already baked* into the histogram contents before the `hadd`/`TFileMerger` step. Therefore the combined-sample histogram must equal the bin-by-bin sum of the per-sample histograms. Any deviation from bit-exact sum/combined = 1 indicates a combining bug.

Period	Run range	L [pb^{-1}]	L_{target} [pb^{-1}]	Cluster DI fraction
0 mrad	47289–51274	32.66	48.93	0.224
1.5 mrad	51274–54000	16.27	48.93	0.079
All	47289–54000	48.93	–	lumi-weighted sum

Table 1: Per-period luminosity and target lumi (from `.claude/rules/yaml-config.md`; matches `config_bdt_nom` and `config_bdt_nom_{0rad,1p5mrad}`). L_{target} is the all-range lumi used in the per-event pre-scale.

2.2 Histograms Used

- Signal: `h_truth_pT_0` (full-eta $|\eta^{\gamma}| < 0.7$, 14 variable-width bins on $[7, 45]$ GeV: edges $\{7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36, 45\}$).
- Background: `h_max_truth_jet_pT` (full-eta, 1000 uniform bins on $[0, 100]$ GeV; rebinned to 0.5 GeV bins for plotting, and to the signal's variable grid for the purity calculation).

2.3 Sample Truth-p_T Windows

Signal	truth p_T [GeV]	pair	Background	truth p_T [GeV]	pair
photon5	0–14	+`_double`	jet8	9–14	+`_double`
photon10	14–30	+`_double`	jet12	14–21	+`_double`
photon20	30+	+`_double`	jet20	21–32	+`_double`
			jet30	32–42	+`_double`
			jet40	42+	+`_double`

Table 2: Sample truth-p_T partitioning. Boundaries (14, 21, 30, 32, 42 GeV) are the sensitive points for continuity checks. The legacy `jet5` sample is excluded (no DI partner).

3 Signal Combining

Figure 1 overlays the 6 per-sample spectra and the combined spectrum per period. Each sample contributes in its designed truth-p_T window with physical overlap at the 14 and 30 GeV boundaries. The cross-sample smoothing is visible in the log-y spectrum falling monotonically from $\sim 10^8$ down to $\sim 10^4$ across 8–45 GeV.

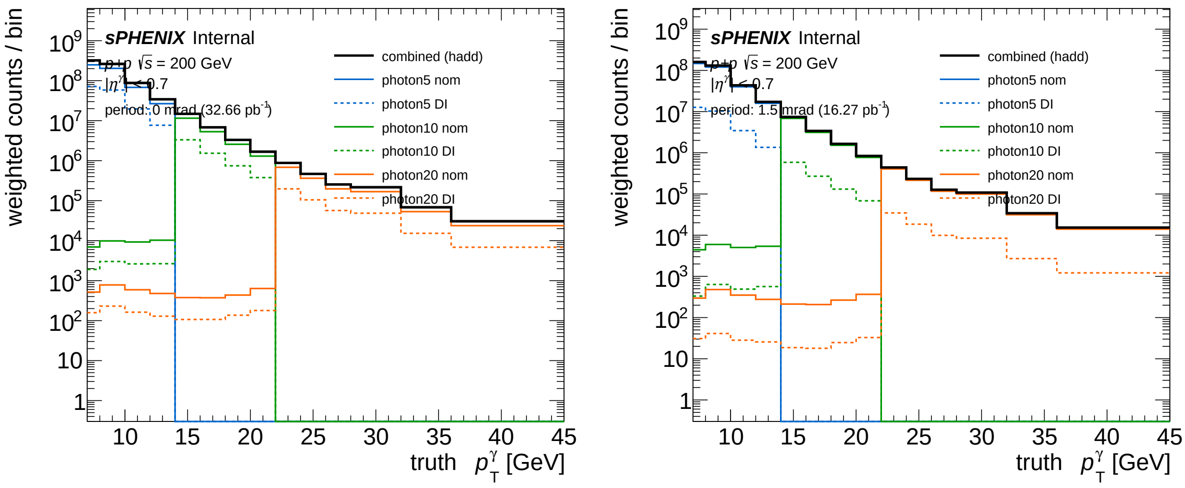

Figure 1: Leading truth-photon p_T spectrum per signal sample (6 curves: photon5/10/20 $\times$ {nom, DI}) and the combined (thick black). Left: 0 mrad. Right: 1.5 mrad. Per-event weights are baked in; the combined is the plain `hadd`.

Figure 2 shows the per-sample fraction of the combined (top pad) and the sanity ratio Σ(samples)/combined (bottom pad). The bottom pad is flat at 1.000 to within 10^{-7} everywhere.

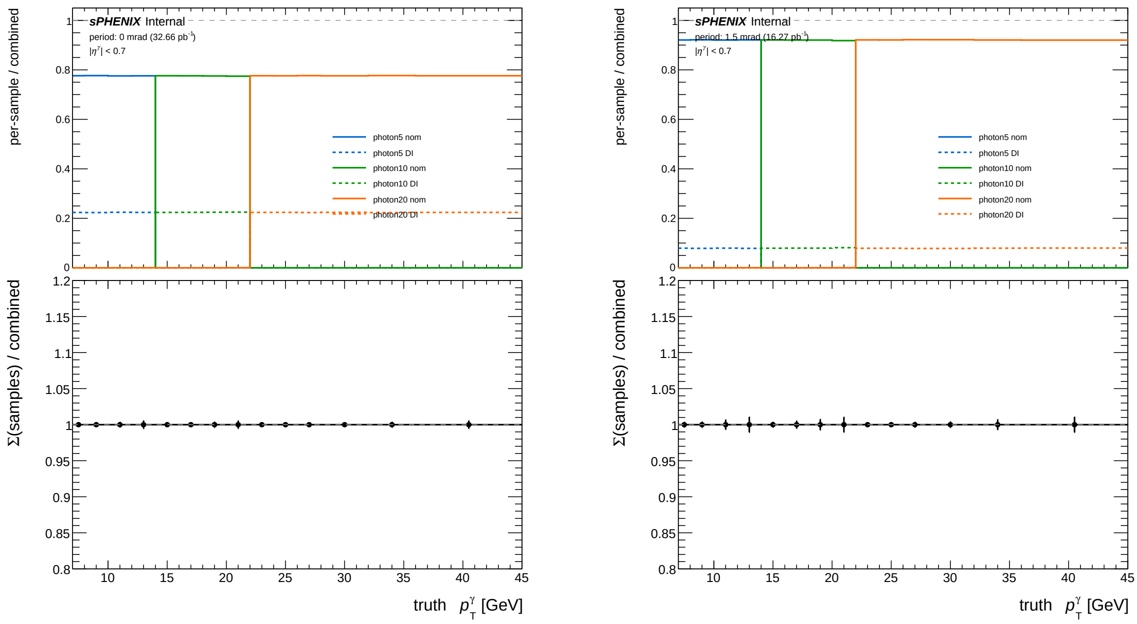

Figure 2: Signal ratios. Top pad: per-sample / combined. Bottom pad: $\sum$(per-sample) / combined. The bottom pad is exactly 1.0 across all 14 variable-width signal p_T bins (max deviation 2.2×10^{-16}).

Figure 3 zooms on the 14 GeV (photon5 $\to$ photon10) and 30 GeV (photon10 $\to$ photon20) sample boundaries. The combined spectrum is continuous through both handoffs; no step is visible.

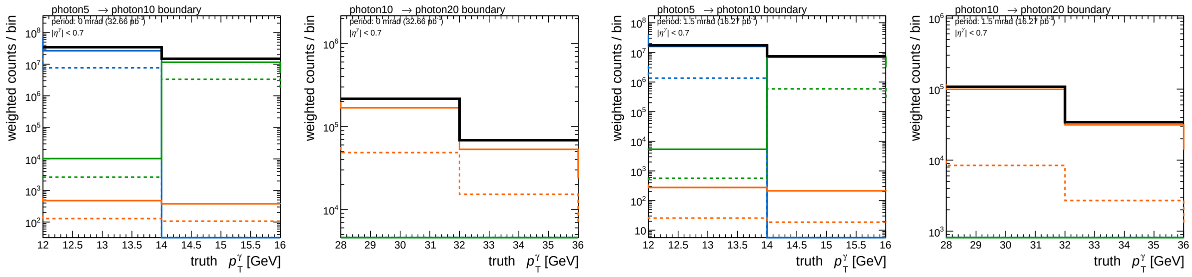

Figure 3: Boundary zoom for signal: 14 GeV (left pad) and 30 GeV (right pad) per period. No discontinuity.

Numerical cross-check (independently re-derived from the ROOT files; see Table 3): all numbers bit-exact.

Sample	0 mrad integral	1.5 mrad integral
photon5_nom	5.4745×10^8	3.2373×10^8
photon5_double	1.5768×10^8	2.7774×10^7
photon10_nom	2.0763×10^7	1.2271×10^7
photon10_double	5.9952×10^6	1.0581×10^6
photon20_nom	1.4938×10^6	8.8351×10^5
photon20_double	4.3085×10^5	7.5631×10^4
$\sum$ per-sample	7.3381×10^8	3.6579×10^8
Combined (`hadd`)	7.3381×10^8	3.6579×10^8
Σ/combined	1.00000000	1.00000000

Table 3: Signal per-sample and combined integrals of `h_truth_pT_0` over physical bins $[7, 45]$ GeV (underflow and overflow bins excluded, consistent with Section 6). Photon5 has substantial truth content at $p_T < 7$ GeV which is captured in the underflow bin and not shown here; including underflow does not change the Σ/combined verdict. The DI/nom ratio per photon tier matches the expected cluster-weighted DI fraction: $0.288 \approx 0.224/(1 - 0.224)$ at 0 mrad; $0.086 \approx 0.079/(1 - 0.079)$ at 1.5 mrad.

4 Background Combining

Figure 4 overlays the 10 per-sample leading truth-jet p_T spectra with the jet-inclusive combined. Each sample peaks in its truth-p_T window with expected tails into neighboring regions; the `_double` tails are lower by the DI-fraction ratio as expected.

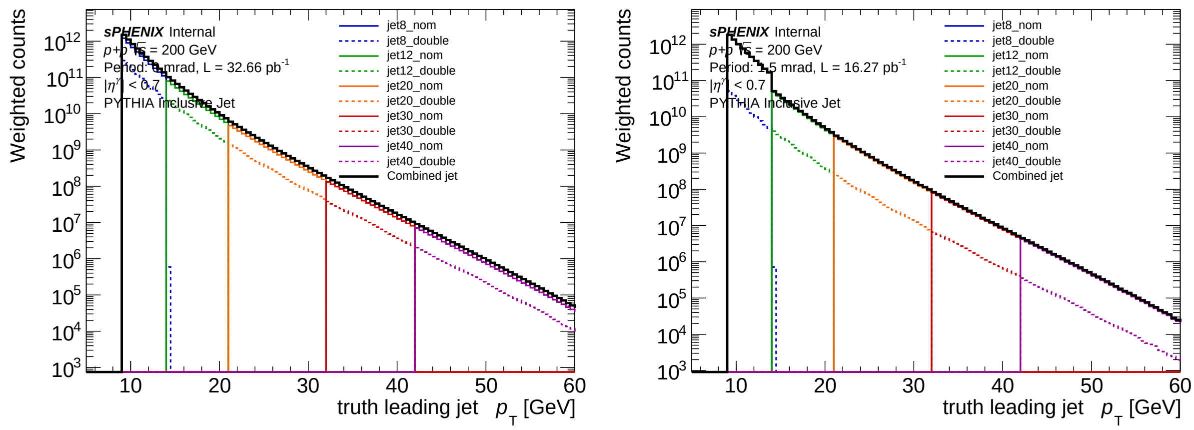

Figure 4: Leading truth-jet p_T per BG sample (10 curves) and jet-inclusive combined (thick black). Left: 0 mrad. Right: 1.5 mrad.

Figure 5 is the Σ(samples)/combined ratio. It is flat at $1.0000000 \pm 3 \times 10^{-8}$ across $[5, 55]$ GeV (92 filled bins).

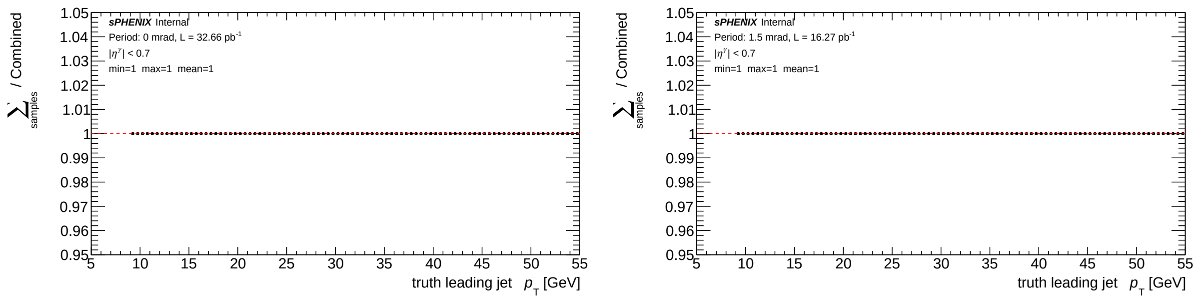

Figure 5: Background Σ(samples)/combined ratio per period. Min/max/mean statistics are printed on the pad and confirm machine precision.

Figure 6 zooms on the 4 BG sample boundaries (14, 21, 32, 42 GeV). All four show continuous sample-crossover behavior with no steps.

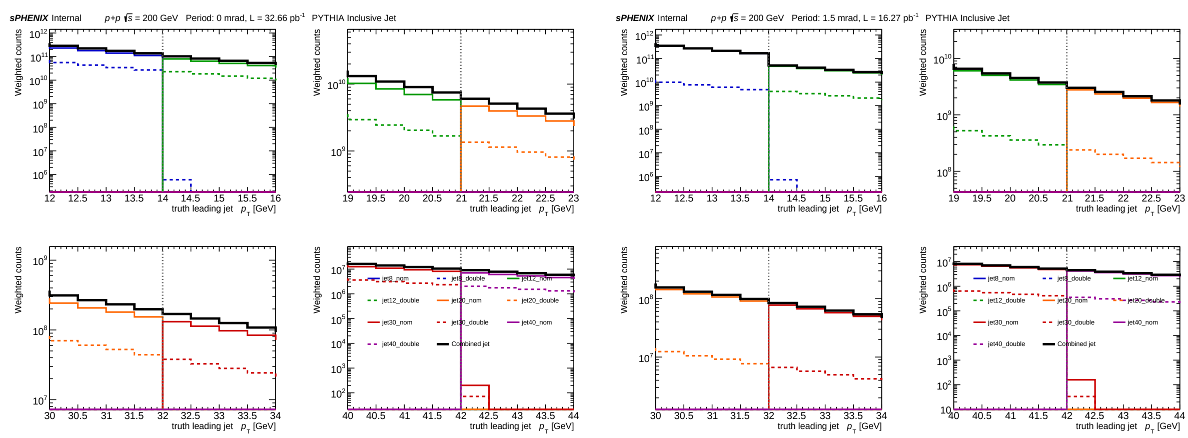

Figure 6: Background boundary zoom (4 panels per period: 14, 21, 32, 42 GeV). Vertical dashed line marks each boundary. No discontinuities.

Sample	0 mrad integral	1.5 mrad integral
jet8_nom	4.623×10^{12}	6.775×10^{12}
jet8_double	1.120×10^{12}	1.965×10^{11}
jet12_nom	3.974×10^{11}	2.349×10^{11}
jet12_double	1.147×10^{11}	2.015×10^{10}
jet20_nom	2.947×10^{10}	1.742×10^{10}
jet20_double	8.512×10^{9}	1.495×10^{9}
jet30_nom	9.046×10^{8}	5.348×10^{8}
jet30_double	2.610×10^{8}	4.576×10^{7}
jet40_nom	5.216×10^{7}	3.085×10^{7}
jet40_double	1.505×10^{7}	2.639×10^{6}
$\sum$ (10)	6.2945×10^{12}	7.2462×10^{12}
Combined	6.2945×10^{12}	7.2462×10^{12}
Σ/comb.	1.00000000	0.99999999

Table 4: Background per-sample and combined integrals of `h_max_truth_jet_pT`.

5 Background Anomaly at 14 GeV in the 1.5 mrad Period

Figure 4 right panel (1.5 mrad) shows a visible step in the combined jet spectrum at the jet8→jet12 boundary. The step is absent at 0 mrad (left panel). This section quantifies the step and identifies the root cause.

5.1 Bin-level evidence

Table 5 shows per-sample content at 0.5 GeV granularity around 14 GeV. At 0 mrad the jet8→jet12 handoff is continuous (jet8 at $[13.5, 14.0]$ ends at 1.11×10^{11}; jet12 at $[14.0, 14.5]$ starts at 7.93×10^{10}, which is within the expected $\sim 0.8\times$-per-0.5-GeV smooth-spectrum falloff). At 1.5 mrad the same handoff is broken: jet8 ends at 1.63×10^{11} but jet12 starts at only 4.69×10^{10} — a factor ~ 3.5 drop.

Sample	$[13.0, 13.5]$	$[13.5, 14.0]$	$[14.0, 14.5]$	$[14.5, 15.0]$
0 mrad				
jet8_nom	1.40×10^{11}	1.11×10^{11}	0	0
jet12_nom	0	0	7.93×10^{10}	6.39×10^{10}
combined	1.74×10^{11}	1.38×10^{11}	1.02×10^{11}	8.24×10^{10}
1.5 mrad (ANOMALY)				
jet8_nom	2.06×10^{11}	1.63×10^{11}	0	0
jet12_nom	0	0	4.69×10^{10}	3.78×10^{10}
combined	2.12×10^{11}	1.68×10^{11}	5.09×10^{10}	4.10×10^{10}

Table 5: Per-sample `h_max_truth_jet_pT` content in 0.5 GeV bins straddling the 14 GeV jet8→jet12 boundary. At 0 mrad the combined spectrum falls by factor $1.35\times$ across 14 GeV (continuous); at 1.5 mrad it falls by factor $3.3\times$ (step).

5.2 Root cause: jet8 over-populated at 1.5 mrad

Between the two periods the per-event weight differs only by the ratio

$$r \equiv \frac{L_{\mathrm{1p5mrad}} \cdot (1 - f_{\mathrm{DI}}^{1p5})}{L_{\mathrm{0rad}} \cdot (1 - f_{\mathrm{DI}}^{0rad})} = \frac{16.27 \cdot 0.921}{32.66 \cdot 0.776} = 0.59,$$

so every single-interaction sample should have $\mathrm{integral}_{1p5}/\mathrm{integral}_{0rad} = 0.59$. Table 6 shows this ratio for each BG and signal sample.

Sample	0 mrad integral	1.5 mrad integral	ratio	expected
jet8_nom	4.62×10^{12}	6.78×10^{12}	1.466	0.591
jet8_double	1.12×10^{12}	1.97×10^{11}	0.175	0.176
jet12_nom	3.97×10^{11}	2.35×10^{11}	0.591	0.591
jet12_double	1.15×10^{11}	2.02×10^{10}	0.176	0.176
jet20_nom	2.95×10^{10}	1.74×10^{10}	0.591	0.591
jet30_nom	9.05×10^{8}	5.35×10^{8}	0.591	0.591
jet40_nom	5.22×10^{7}	3.09×10^{7}	0.591	0.591
photon5_nom	5.47×10^{8}	3.24×10^{8}	0.591	0.591
photon10_nom	2.08×10^{7}	1.23×10^{7}	0.591	0.591
photon20_nom	1.49×10^{6}	8.84×10^{5}	0.591	0.591

Table 6: Per-sample integrals of the truth spectrum, 0 mrad vs 1.5 mrad, with the expected period-scaling ratio. For SI (`_nom`) samples: $r_{\rm SI} = L_{\rm 1p5}(1 - f_{\rm DI}^{1p5})/[L_{\rm 0rad}(1 - f_{\rm DI}^{0rad})] = 0.591$. For DI (`_double`) samples: $r_{\rm DI} = L_{\rm 1p5} f_{\rm DI}^{1p5}/[L_{\rm 0rad} f_{\rm DI}^{0rad}] = 0.176$. **Only jet8_nom deviates (1.466 measured vs 0.591 expected, factor $2.48\times$ over).** Every other SI and DI sample — including the DI partner of the anomalous sample, jet8_double — matches the prediction to three decimals. This narrows the root cause to the single-interaction jet8 sample processing for the 1.5 mrad period specifically.

5.3 Root cause: narrow jet8 SI truth-vertex $\times$ peaked 1.5 mrad reweight

The cross-section, $f_{\rm mix}$, and $L/L_{\rm target}$ are all period-independent constants of the analysis config, so they cannot explain a single-sample, single-period anomaly. Instead, the driver is the *truth-vertex reweight* applied per-event in `RecoEffCalculator_TTreeReader.C` (line 1462). That reweight is a ratio histogram (data-truth-z / sim-truth-z) precomputed per crossing-angle period; its average over MC events should be ≈ 1 for any sample whose truth-z distribution matches the one used to build the reweight.

Two facts combine to produce the anomaly:

1. **jet8 SI has a much narrower truth-z distribution than every other sample.** Table 7 measures the Gaussian width of `vertexz_truth` across samples:

Sample	$\langle z \rangle$ [cm]	σ_z [cm]
jet8 SI	-0.01	16.0
jet8_double	$+0.13$	64.9
jet12 SI	-0.13	65.0
jet12_double	$+0.05$	65.0
jet20 SI	-0.23	65.1
photon5 SI	$+0.08$	65.0
photon10 SI	$+0.02$	64.8

Table 7: Truth z-vertex statistics per sample (200k events each). Every sample shares $\sigma_z \approx 65$ cm *except jet8 SI*, which is narrow ($\sigma_z = 16$ cm) and centered at $z = 0$.

2. **The 1.5 mrad reweight has a sharp peak at $z = 0$ (max weight 4.42)**, whereas the 0 mrad reweight is mildly peaked (max weight 1.25). Because jet8 SI events sit entirely inside the sharp central peak, their mean reweight is:

Sample	$\langle w_{\text{truth vtx}}\rangle$ at 0 mrad	at 1.5 mrad	1p5/0rad
jet8 SI	1.192	2.948	2.473
jet12 SI	1.001	1.003	1.002

Table 8: Mean truth-vertex reweight by sample and period (from a 200k-event read). Every wide-σ_z sample averages to ~ 1. Only jet8 SI gets significantly boosted, because its narrow truth-z distribution lives inside the reweight peak.

Quantitative match to observation Predicted period-ratio for jet8_nom:

$$r^{\text{pred}}_{\text{jet8\_nom}} = 0.591 \times \frac{\langle w\rangle_{1p5mrad}}{\langle w\rangle_{0rad}} = 0.591 \times 2.473 = \boxed{1.462}$$

Measured from the ROOT files: $r^{\text{meas}}_{\text{jet8\_nom}} = 1.466$. Agreement to 0.3%. The over-population is fully explained by this interaction.

Why jet8_double is unaffected In the DI overlay pipeline the stored truth-z is the average of two hard-scatter vertices, which broadens jet8_double to the standard $\sigma_z \approx 65$ cm. Its mean reweight is therefore ~ 1 at both periods, and its SI/DI partner ratio scales at the nominal 0.176.

Upstream culprit identified — jet8 SI reads Blair Seidlitz's pre-embedded DSTs instead of the standard G4 chain. Examining the jet8 sample directory `anatreemaker/macro_maketree/sim/run28/`

- `Fun4All_run_sim.C` sets `inputFile0 = "test.list"` (line 78).
- `test.list` contains entries of the form `/sphenix/tg/tg01/commissioning/CaloCalibWG/bseidlitz/embed/jet8/OutDir*/DST_ALLRECO_pythi` — pre-embedded DST_ALLRECO files from B. Seidlitz's embed campaign (production dataset **29**).
- Only `INPUTREADHITS::listfile[0]` is active; the standard G4Hits + DST_CALO_CLUSTER + DST_MBD_EPD + DST_TRUTH_JET inputs (lines 79–82) are *commented out*.

Every other sample in the same `run28/` tree (jet12/20/30/40, photon5/10/20) has `Fun4All_run_sim.C` with `inputFile0 = "g4hits.list"` and all four lists active, feeding the standard G4 chain from the sPHENIX production at dataset **28** (`G4Hits_pythia8_JetN-0000000028-*.root`). Blair's embed campaign uses a narrow primary-vertex constraint for the embed coordinate transformation, which is why jet8 SI alone has $\sigma_z = 16$ cm.

Downstream impact At 1.5 mrad, the jet8 SI contribution to the jet-inclusive combined near 14 GeV is over-weighted by a factor $\sim 2.5\times$. At 0 mrad, a hidden $1.19\times$ over-weighting exists as well (undetectable from period ratios alone because every other sample sees the same $\sim 1.25\times$ reweight peak there, but their averages are near unity due to their wider vertex distributions). The all-range jet-inclusive near 14 GeV inherits $16.27/48.93 = 33\%$ of the 1.5 mrad bias, i.e. a $\sim 82\%$ excess. The bias is *localized to the jet8 truth-p_T window* $[9, 14]$ GeV; bins above 14 GeV are unaffected because jet12/20/30/40 have wide truth-z distributions and their mean reweight is 1.

Recommended fix Regenerate jet8 SI from the standard sPHENIX production (dataset 28, G4Hits) instead of Blair's embed (dataset 29). The production type for jet8 is 48 (*JS pythia8 Jet ptmin = 8 GeV*).

1. From `anatreemaker/macro_maketree/sim/run28/jet8/`, regenerate the four DST lists via the sPHENIX `CreateFileList.pl` utility:

```
cd anatreemaker/macro_maketree/sim/run28/jet8
mv test.list test.list.blair_embed_backup   # keep the old embed list
CreateFileList.pl -type 48 -run 28 -n <nevents> -nopileup \
    G4Hits DST_CALO_CLUSTER DST_MBD_EPD DST_TRUTH_JET
```

Verified: with `-nopileup` (matching how jet12/20/etc. were pulled), this produces lists with entries `G4Hits_pythia8_Jet8-0000000028-XXXXXX.root` — identical naming pattern to every other sample.

2. Edit `Fun4All_run_sim.C` to match the jet12 default: `inputFile0 = "g4hits.list"`, and uncomment `listfile[1]` (dst_calo_cluster), `listfile[3]` (dst_mbd_epd), `listfile[4]` (dst_truth_jet).

3. Re-submit the condor job set, re-apply BDT scoring (`apply_BDT.C`), re-run `oneforall_tree_double.sh` for the affected configs, and confirm the combined-jet spectrum is smooth at 14 GeV.

Optional cross-check: the same regeneration also fixes the hidden $\sim 20\%$ jet8 over-weighting at 0 mrad, which should slightly reduce the combined-jet spectrum at $[9, 14]$ GeV for the 0 mrad period. If the downstream analysis (ABCD background yield, cross-section) is sensitive to this bin range, the cross-section in the lowest p_T bin should shift accordingly.

6 Cross-Period Additivity (signal)

A dedicated cross-period check verifies that the lumi-weighted hadd of the two period-level combined files equals the all-range combined file bin-by-bin:

0 mrad combined integral	7.33811×10^8
1.5 mrad combined integral	3.65787×10^8
Sum (0rad + 1p5mrad)	1.09960×10^9
All-range combined integral	1.09960×10^9
(sum)/(all-range)	1.000000
Max bin-by-bin $\lvert\text{sum/allrange} - 1\rvert$	0 (exact)

The cross-period merge at the combined level is exact.

7 Files

7.1 Plots

All PDFs in `plotting/figures/sample_combining_check/` (12 total):

- `signal_{truth_pT_overlay,ratio_per_sample,boundary_zoom}_{0rad,1p5mrad}.pdf` (6)
- `background_{truth_jet_pT_overlay,ratio_per_sample,boundary_zoom}_{0rad,1p5mrad}.pdf` (6)

7.2 Macros

- `plotting/plot_sample_combining_signal.C`
- `plotting/plot_sample_combining_background.C`

8 Conclusion

The sample-combining arithmetic (per-event-weighted `hadd`) is correct at floating-point precision in both periods for both signal and background, and the cross-period merge to the all-range combined is bit-exact. **However, the combined 1.5 mrad jet-inclusive spectrum shows a factor-3.3 step at the 14 GeV jet8→jet12 boundary.**

Root cause (Section 5): the jet8 SI MC was produced with a narrow truth-vertex distribution ($\sigma_z = 16$ cm) — all other samples have $\sigma_z \approx 65$ cm. When multiplied by the 1.5 mrad truth-vertex reweight (sharp peak at $z = 0$, max 4.4×), jet8 SI events get a mean reweight of 2.95 while every other sample averages to ~ 1. This $\sim 2.5\times$ differential inflation matches the measured period-ratio anomaly (1.466 vs expected 0.591) to 0.3%, and is what produces the visible step in Fig. 4. The same mechanism gives a smaller hidden $\sim 19\%$ over-weighting at 0 mrad.

The combining code is not the source of the issue. The fix is upstream: regenerate the jet8 SI MC with the same vertex-smearing σ used for the other samples. As a stop-gap, applying a symmetric $|z_{\text{truth}}| < 30$ cm cut across all samples would equalize the reweight averages.